Semiconductor & Infrastructure Software Industry – Arista Networks, Astera Labs & Fabrinet – Senior Architect at Nvidia Corp

Consultant | 16 October 2025

Specialist Background

- Around 20 years' experience in the semiconductor industry, focusing on mobile chips, custom accelerators, cloud systems and data centre infrastructure
- Well-versed in the semiconductor and infrastructure software industry's competitive landscape, including power chips and subcomponents companies
- Well-placed to discuss whether Arista will lose share because of advancements being made on the Spectrum-X, as well as Astera Labs' growth runway and where Fabrinet sits in the supply chain
- Knowledgeable on how far we are on the Chip-on-Wafer to eliminate the substrate, as well as cold plating and HBM (high-bandwidth memory) technology

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Analyst:
We're looking to better understand the semiconductor and infrastructure software industry. We're familiar with Google TPU [tensor processing unit] 7, Trainium and AMD [Advanced Micro Devices] is coming up with the MI350, 450 in 2026. Do you believe these chips are close to competing with Nvidia?

Specialist:
In general, I don't think any of these companies are close. You talked about the TPU, I would probably put it around six on a scale of 10. Then MI450, again, I would put it around six or seven. Then Trainium, again, I would put it at six.

Analyst:
I know of Groq and Cerebras, and there are a few others that are coming up.

Specialist:
Yes. No, they are not, and they are struggling. I have friends at these companies. It's hard for them to find customers that can meaningfully add significant revenue. Also, the way to think about this is also like what kind of workloads you are attacking. We are talking about like inference space. Of course, in training space, I don't think all of these competitors are much farther behind. In inference space, like MI450 is definitely a good contender. I still don't think that they are where they need to be in terms of software yet.

Then the TPU is also, I would say, a pretty good reasonable contender in the custom accelerator space, but also keeping in mind that it's mostly for specific workloads like internal workloads that Google optimises the TPU for something like YouTube recommendations or Gemini. This is not something very surprising. GPUs are what they have been since a while. They are general purpose accelerators, still accelerators, but not suited for any specific workload. They perform well at most of the workloads. Then you have the specialised architectures that are basically tailored or custom-made for transformer architecture.

You have different variants like you mentioned about Groq and Cerebras, they want to take advantage of almost infinite amount of memory available to them, which again works well for a lot of workloads. For a majority of the workloads, they are still not competitive, I would say, in real life. Software is one piece that I mentioned, but also networking is the other piece. When you think of the new unit of computing, that's the data centre as a whole. You have to think in the scale of like rack-level computing rather than like a single chip. You have the chip, you have the connectivity part, networking, again, Nvidia is far ahead than some of the alternatives like Ethernet for scale-out or PCIe for which most of these other companies use.

Analyst:

You mentioned the whole rack level and the networking lens as well. What are the big architectural changes happening? Do you believe a company such as Arista will lose share because of the advancements being made on the Spectrum-X?

Specialist:
Yes. No, it's a good point. I know that the opinion was that Arista Net is going to lose. I don't really think so because Arista Net is a competitor to InfiniBand and Spectrum-X, but their solution is very different. Again, along with the Broadcom, they have the 400G, 800G, which is like the vanilla out-of-the-box network switching solutions and connectivity solution that just works. Some of the hyperscalers like AWS don't like Nvidia connectivity because they don't want to pay licensing fees. They don't want to pay the margins and so on and so forth. They have their own version of Ethernet, but they also use to an extent, the Broadcom switches. They also use to an extent, Arista. At the same time, Meta also in some of their data centres, it continues to use Arista and Broadcom. Google, of course, I don't know if they use Arista for the network connectivity. They might have some of their own connectivity solution itself.

I think Arista also has more room to grow, like someone like Google or Amazon could also basically adopt their switching solutions. I know that other than AI data centres or servers itself, like for HPC, Arista is like a very, very popular name that's being used.

Analyst:
When we look at component companies, there are people building power chips and other subcomponents. Which companies do you believe have the best solution?

Specialist:
We are still talking about the networking side? Is it like scale-up network?

Analyst:
The scale-up side, because it seems like you're trying to build bigger and bigger solutions.

Specialist:
Yes. I think one is definitely one that has established scale-up solution is Google, which is using their own interconnect solution in their TPUs for scale-up solutions. The other one, which is, again, like I was talking about the UALink, it's not adopted, but AMD, Intel, and Broadcom are basically trying to adopt that and commercialise that more. I think the most promising right now is the PCIe gen 5 and gen 6 that most of the hyperscalers, they are using some version of this, and it's also compatible with the standard switch fabrics from Broadcom. It doesn't need a lot of cost-sensitive hardware and PCIe is basically just open-source protocol. I think that is the most widely adopted right now. Now in terms of the future, it's hard to say because with NVLink licensing opportunity,

Analyst:
Who builds the UALink [Ultra Accelerator Link]? Is there another company that can benefit from it?

Specialist:
Yes. No, there is no one company. It's a consortium of multiple companies like it's backed by AMD, Intel, Broadcom, Cisco, Google, and Microsoft, Meta, all of these companies, they are backed it. It's not gotten traction up until now.

Analyst:
You mentioned gen 5 and 6. Is that also proprietary internal, or is there an external company that offers that?

Specialist:
The PCIe gen 5, gen 6, those are open-source protocols. They are open-source protocols, but say you want to implement that, then companies like Synopsys would provide the SerDes for that for you to integrate as an IP. Then you need retimers for PCIe. Companies like Astera Labs are the leaders that support those.

Analyst:
Let's move to semi and component companies, such as Astera Labs, that have really amazing technology. The company seems to be doing a little on the active cable. Which other company comes to your mind?

Specialist:
Yes, Astera Labs and there is another company called Credo. Again, those are the two leaders. I would say those are the two legit companies that would come to my mind, especially for the connectivity, like AEC connectivity. Now there are a bunch of other companies that are also doing optical connectivity and design optical transceivers. Then there are even fewer companies which are basically working on integrating the optical photonics on the board and on the package. It's silicon photonics and the technology is called CPO, co-packaged optics.

Analyst:
Do you believe Astera Labs and Credo have a long runway? How much room do they have?

Specialist:
Correct. Yes. I think they should still have a pretty long runway because copper cabling is not going to go anywhere. Say, it's at 100% today, say, five years from now, it will be probably at still 80-90% purely because it's just cheaper and with active electrical cables also have their own signal integrity issues. For your rack-scale connectivity, you don't need to travel several yards or several metres. They are cheaper, they are reliable, so they should continue to work. For scale out, definitely, you need to look out for these other alternatives because currently, they are more expensive.

Analyst:
Are you familiar with SiTime? Do you know about the timing chips?

Specialist:
I have heard about it, I have heard about SiTime, but I don't know a lot about how the performance is or who is using them. I believe probably some hyperscalers is already using them.

Analyst:
You are not using it for doing the timing chips, right?

Specialist:
Right. Yes. No. For PCIe retimers, Astera Labs is pretty much the de facto.

Analyst:
We're also interested to learn about Fabrinet on the assembly side, and it seems Nvidia and Mellanox has a pretty tight relationship with them.

Specialist:
Right. Yes.

Analyst:
Where does Fabrinet sit in the supply chain, and if you aren't using it for the assembly, which player does that step go to? Who are the competing vendors?

Specialist:
Fabrinet comes like pretty, I would say, like late in the design cycle, like it's more at the subsystem and module assembly chain. It sits more in the optical transceivers and co-packaged optics assembly. It's like the completed PCB assembly enclosure, system level burn-in. Those would be some of the last major steps, at the PCB level itself on the assembly build line.

Analyst:
Given you don't use Fabrinet for the module assembly, what other companies accomplish that for you?

Specialist:
There is Foxconn for assembly. Then there are other alternatives. There is Coherent Corporation, there is Jabil. There is also like AWS, we also had InnoLight. Those are some major ones.

Analyst:
Which companies do you believe have a lot of growth ahead of them? We know HPM [Hanwha Precision Machinery] is getting very popular, but I'm sure there are others we don't know of. Are there other components that are becoming more critical, as complexity is increasing?

Specialist:
Yes. The cold plate component is definitely something to watch out for. You have also just the components that are involved in the high-speed interfaces, connectors. We already talked about retimers. One component we didn't talk about is power management modules like VRM. Infineon is one company that makes them. There is also MPS. I would look at these two companies because these are pretty much in high demand, even though they are not like a very high-margin business, but just a volume point of view, I would be looking at those. You mentioned about HBM memory. There are also other memories, so memory makers itself like Micron and SK Hynix. Even though not every server would use HBM, so you could still have LPDDR or GDDR memory like low-cost memory, NAND, NVMe components. Then we talked about the advanced interconnects, which have a very tight integration with HBMs, with the GPUs, AI accelerators. These are like high-density interconnect PCBs. We are talking about Microvias, more finer wire traces, stacked designs.

There is also a technology which I believe could become big after CoWoS. It's called Chip-on-Wafer-on-PCB. You eliminate the substrate part and you directly are basically using the PCB as your substrate layer as a connectivity between different chiplets. That's promising. I mentioned all of these, I would say are right.

Analyst:
How far are we on the Chip-on-Wafer to eliminate the substrate? Who makes that?

Specialist:
CoWoP, I would say it is in a pretty advanced stage. We could see production side of it like sometimes maybe next year or the year after that. It is in a pretty advanced stage, and it's like developed as an alternative to all the supply chain bottlenecks that we are seeing with CoWoS itself. As to who makes that, again, TSMC has the technology, has the capability to make it. Samsung has the capability to make true 3D packaging with interposer and PCB integration. Amkor also has that capability. Yes, I think probably Unimicron is another company. ASE is another one. All these are like they have the technology to make this.

Analyst:
On the power management side, you mentioned MPS. Is that Monolithic Power?

Specialist:
Yes. MPS is Monolithic Power Systems.

Analyst:
Is there any alternative to them or is Monolithic the best in technology?

Specialist:
Yes. I think both Monolithic and Infineon, I would say they are both pretty good. Monolithic is smaller. That's why I would probably give them one extra point if I had to rate them on a scale of 10 compared to Infineon, their support, their technology is still pretty good for a small company like that.

Analyst:
Who makes the advanced interconnects with HBM [high-bandwidth memory] and high-density interconnects?

Specialist:
I would say pretty much the same players who are usual suspects for PCBs.

Analyst:
Power management must be quite critical. As power requirements increase, does the amount of content Monolithic Power Systems has on a single board also increase?

Specialist:
Yes, definitely because previously, we had one or two phases that would deliver the power to the entire package or the chip. Then we had like eight phases, then we moved to 16, 32 phases. You need pretty sophisticated power management and VRM modules to handle that. At the same time, you also have a new technology called backside power delivery by which traditionally, you use the same metal layers within a chip for the power delivery, and so they have to fight with the resources on the wiring and connectivity side within a chip. With the backside power delivery now, you can supply the power to the chip from one side, whereas you could use the other side for all your interchip routing resources. That technology is also pretty new. That also has additional complications. I would say the amount of content on the PCB is increasing for someone like MPS to handle and get more share.

Analyst:
You mentioned the cold plating technology. Which player does that?

Specialist:
I think there is Vertiv, who is the leader. There is Supermicro, who has their own liquid cooling. There are others like Schneider who also have the liquid cooling technology, but Vertiv is the leader.

Analyst:
On the component side, they make bigger systems for data centre cooling, but I was thinking about the chip level or the board level cold plates. Are you familiar with a company called Asia Vital Components? It does thermal cooling and makes a lot of components for liquid cooling for the chip. It used to do the cooling for notebooks.

Specialist:

Probably not. I've heard about Thermal Dynamics, and I've heard about Boyd.

Analyst:
You mentioned believing TPUs will probably still take a little while to compete, at least a few more years. Do you think it can come close to inference in 2027?

Specialist:
Yes. If you want to do something like a public cloud service, I think it will still be longer than that because today, they are purely optimised for the internal workloads. When you open up your infrastructure to the outside world, you need to ensure that it works seamlessly with each and every workload. That's the point that I was trying to allude that these are optimised for very, very specialised workloads. It will take longer than just maybe one year or two to make them work for each and every workload.

If, say, Google were to open up the TPU ecosystem to the outside world, which I'm sure they will sooner or later, how much time will it take for someone like a small start-up who doesn't probably know anything about their TPUs to get on the TPUs and start using them, something which we are seeing with AMD. They struggle pretty much for 2-3 years to even come up with a strategy for their software, even though their hardware is pretty competitive.

Transcription ends